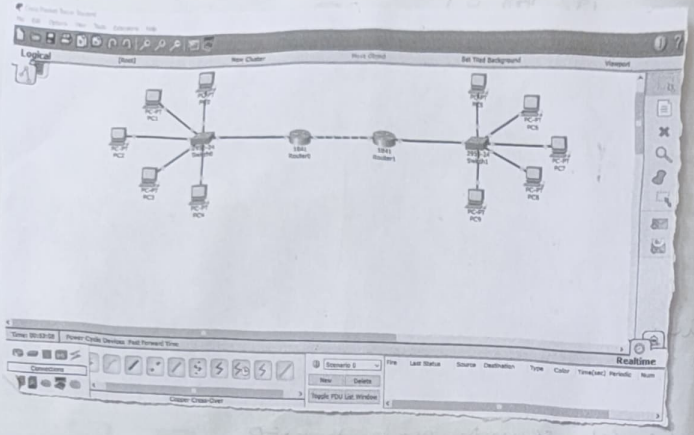
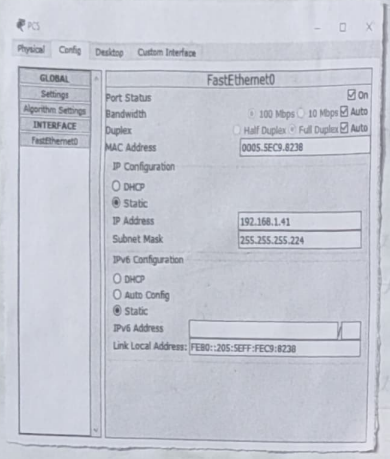
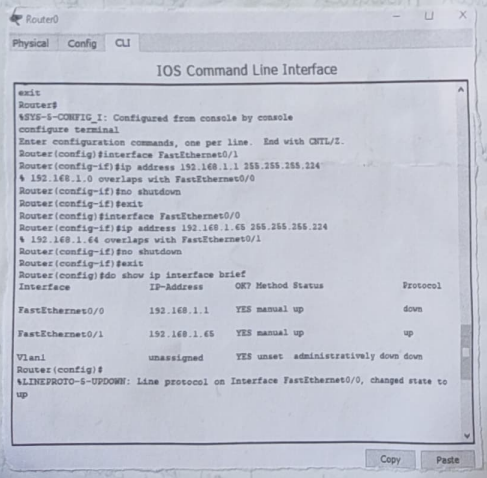
Ex 9 IMPLEMENTATION OF SUBNETTING IN CISCO PACKET TRACER SIMULATOR
27/8/26





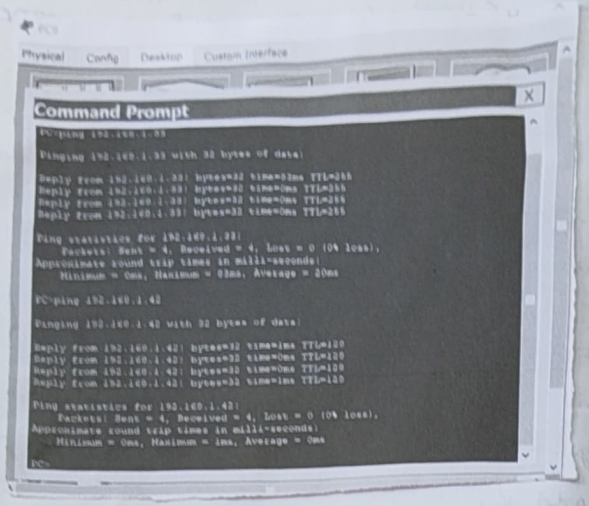


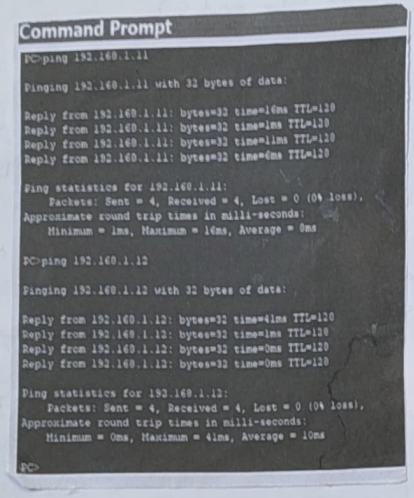
AIM:

To implement classless IP subnetting using Cisco Packet Tracer and configure the network devices with appropriate IP addresses and subnet masks to establish communication between different subnets

PROCEDURE:

* In Cisco Packet Tracer create a new network topology
* Add the required network devices:
  - ⇒ 2 routers (R0 and R1)
  - ⇒ 2 switches (S0 and S1)
  - ⇒ 10 PC's (PC0 - PC9)
* Connect PC0 - PC4 to switch S0 and PC5 - PC9 to switch S1 using suitable ethernet cables
* Connect R0 and R1 using available fast Ethernet interfaces.
* Divide network to /27 subnets
* Enable router interfaces using the no shutdown command
* Configure R1 using CLI. Assign 192.168.1.33 to interface connected to S1 and 192.168.1.66 to the interface connected to R0.
* Enable the R1 infaces using no shutdown command in end





(iii). Configure the IP address of PC0-PC4 in 1st subnet

(a) PC0 - 192.168.1.11
(b) PC1 - 192.168.1.12
(c) PC2 - 192.168.1.13
(d) PC3 - 192.168.1.14
(e) PC4 - 192.168.1.15

use subnet mask 255.255.255.224 and default gateway 192.168.1.1

(iv) Configure IP address of PC5-PC9 in the 2nd subnet

(a) PC5 - 192.168.1.41
(b) PC6 - 192.168.1.42
(c) PC7 - 192.168.1.43
(d) PC8 - 192.168.1.44
(e) PC9 - 192.168.1.45

Use subnet mask 255.255.255.248 and default gateway 192.168.1.33

(v) configure static routes on R0 and R1 so that packets can travel b/w 2 LAN subnets

(vi) Verify the router interface using commands

(vii) If successfully replies are received with no packet loss, the submitted network is functioning correctly

RESULT:

Thus classless IP subnetting was successfully implemented using Cisco Packet Tracer and connectivity between the subnetted networks was verified using Ping Command